

Individual Assessment Coversheet

To be attached to the front of the assessment.

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Indicate	Yes	No
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Section A

Question 1

1.1) Essential security principles that should be taken into consideration is ensuring that the confidentiality, integrity and availability of data for clients and the business in the network. Data confidentiality refers to making sensitive information only accessible to authorised individuals. Data integrity is ensuring that the data in the network remains accurate by preventing unauthorised individuals from maliciously altering it. Data availability refers to ensuring that the data in the network is accessible to users such as employees and clients.

In addition, security features such as firewalls, data encryption and secure access control are essential for ensuring that the network is secure and can be scaled more multiple devices in the business. Implementing and configuring a firewall for hardware and software can assist in preventing malicious traffic by restricting traffic that occurs outside of the business. Moreover, utilising protocols like the transport layer security protocol enables the business to encrypt data when it is being transmitted to prevent sensitive data from being collected and tampered with. Furthermore, there are various methods that can be utilised in order to integrate access control to the network such as implementing two-factor authentication when accessing accounts as well as implementing network access control that enables the business to regulate the network by creating predefined rules that ensures that only authorised users and devices to connect to it (University of San Diego, 2024).

1.2) Integrating cloud services to assist in the expansion of the network can assist in scalability and flexibility as it enables the business to easily add computing resources through virtual server instances to meet any network demands without the company needing to utilise additional servers. Moreover, firewalls can be configured in a manner that allows it to automatically scale with the network and provide security for the whole network. Additionally, network access control proves to be beneficial in terms of scalability, because the predefined rules that they follow will always remain consistent and can automatically scale to the number of devices that are using the network. However, using the transport security layer protocol can prove to be difficult to make it scalable for the network as it will require additional computing resources. Methods like load balancing and caching through the cloud can be used to support the increase in traffic (PubNub, 2024).

Implementing ethernet switches to the infrastructure to the network can assist in increasing overall performance and security. A switch can come with hundreds of ports allowing for a large number of devices to connect to it, which can prove to be beneficial for scalable networks. In addition, each port on the switch has its own collision domain that greatly reduces the number of possible collisions that can occur over the network and greatly improves network performance. Switches also utilise MAC addresses to transfer data packets directly to the correct device and allows for full-duplex communication, enabling data to be transferred and received simultaneously, that further improves efficiency and performance.

Integrating and configuring a VLAN on these switches can further improve network performance as well as security. VLANs segment the switch into smaller switches allowing for more logical networks without additional hardware making the network more scalable. It can also make the

network secure as it can separating important areas from each other and limiting access to reduce the risk of unauthorised access to sensitive data. Separating network areas from each other can also improve performance since traffic is also separated from each other.

Question 2

2.1a) Internal vulnerabilities that can occur in CYBER Solutions can include providing unnecessary permissions to individuals in the business such as employees and business partners that can act as an insider threat. Moreover, utilising legacy software systems or not updating a system software to the most recent security patch can lead to various vulnerabilities that can be exploited by external threat actors to perform cyber-attacks like spreading malware and stealing sensitive data. In addition, having poor physical security of network infrastructure can lead to external threat actors to commit cyber-attacks by infiltrating the business premises and easily gaining access to network hardware.

External threats can include external threat actors conducting cyber-attacks against the business's network remotely through methods like DDoS and phishing attacks. Furthermore, external threat actors are also able to use social engineering techniques with the business's employees in order to spread malware that enables them to gain access to the network.

2.1b) External vulnerabilities like external threat actors can perform cyber-attacks that can greatly diminish the security and stability of the business's network. Examples of possible cyber-attacks that can be carried out onto CYBER Solutions can include DDoS attacks that prevents employees and clients from accessing data over the network, because the aim of this cyber-attack is disrupt the network traffic and stability. Additionally, malware such as spyware can be developed and spread to the business's network for the purposes of gathering sensitive data from employees and clients, which weakens the data confidentiality of the network. Moreover, phishing attacks can also be conducted in order to gain access to the network where the external threat actors can then steal confidential data or damage the business operations and stability by diminishing the integrity of important data for clients and the company. Furthermore, man-in-the-middle attacks can be performed to further weaken data confidentiality by intercepting data between the business and clients. In addition, external threat actors can utilise methods like social engineering and malware to gain unauthorised access to the business network where they are able to easily steal sensitive business and client information as well as tamper with the data and damage the network itself.

2.2) Utilising log reviews can be used in identify vulnerabilities by monitoring and analysing network traffic and protocols. Potential areas where vulnerabilities can be found are indicated by any trends that are identified through the log reviews.

After determining the potential areas where network vulnerabilities are located, a vulnerability assessment can then be conducted to determine if the current configuration meets the ideal configuration for the business. This can be accomplished with automated vulnerability scanners that can perform tests using the Common Vulnerabilities and Exposures (CVE) dictionary to identify any vulnerabilities in the system. By identifying any common vulnerabilities, CYBER Solutions are then equipped with the knowledge on how to mitigate the vulnerabilities that were detected, which increases the overall security of the network. Furthermore, vulnerabilities that can

be exploited by external threat actors to gain access to the network, steal as well as tamper with data, like firewall misconfiguration and unsecured networks, should take priority over other vulnerabilities that were discovered.

After identifying and implementing solutions for all the vulnerabilities found, penetration tests should be conducted to ensure that the network is fully secure. Penetration accomplishes this by utilising hacking techniques to determine any additional weaknesses in the network by exploiting vulnerabilities. Performing this test can indicate new vulnerabilities that were not discovered during vulnerability scanning and the business can then develop processes that can mitigate these vulnerabilities and increase network security. The most devastating vulnerabilities, such as gaining unauthorised access to the network, should be tested first when performing a penetration test to ensure that the network is protected against the most damaging cyber-attacks.

Question 3

3.1a) Employing training programmes are a type of prevention-based security control. This type of programmes enable organisations to greatly improve their staff's knowledge over various cyber-attacks and how to effectively manage sensitive company data. Therefore, employees will become more equipped with handling sensitive information in a manner that does not make it vulnerable to potential cyber-attacks as well as how to identify and avoid instances of cybersecurity threats and techniques like social engineering.

3.1b) Training employees to recognise instances of phishing will prevent them from revealing sensitive company information, like account passwords, or download malware. The severity of falling for a phishing attack is dependent on what information cybercriminals want to extract from unknowing employees or what they want the employee to perform that will provide them an advantage. Utilising strong and unique passwords for each system that requires one will make it more difficult for cybercriminals to break into these company systems, which helps to prevent potential security breaches. Moreover, incorporating safe internet habits such as enabling two-factor authentication, avoid suspicious links and using secure websites can help to further prevent potential security breaches, because it makes it more difficult for cybercriminals to gain access to sensitive information through tactics like spreading malware.

3.1c) Conducting a training session that educates employees on all the different types of cyber-attacks within the industry as well as they can be identified and its potential impact will prove to be beneficial for CYBER Solutions, because employees will have a better understanding of the severity of different cyber-attacks and will be more likely to actively avoid them. Moreover, explaining how the different security policies in the business should be followed ensures that employees fully understand what is expected from them to make sure that sensitive information stays secure. Additionally, conducting training sessions on how to apply security measures to prevent the possibility for cyber-attacks to be successfully can also be beneficial to the business as employees are more likely to implement their own security procedures to further ensure that the business is protected from potential cyber-attacks.

3.2a) Creating a security-conscious culture within the business can introduce various benefits that can enhance overall security can include having a workforce that is more engaged and compliant with security measures and procedures. Additionally, it leads to a decrease in the risk for security vulnerabilities and threats to be done by insiders as they will be more knowledgeable about potential threats. Furthermore, employees will also be more likely to stay up to date with any recent potential security threats and be prepared to prevent it from occurring (National Protective Security Authority, 2023). Moreover, employees can identify other employees who are accidentally creating security vulnerabilities and them in effectively following security procedures. However, if the employee is creating security vulnerabilities on purpose, then the other employees will be more likely to report them to management.

3.2b) The effectiveness of cyber awareness training in a business can be measured through various methods such as reduced risky behaviour from employees that can introduces vulnerabilities to network and improved decision-making with regards to experiencing unsecure situations that can occur (Sunil Chaudhary, 2022). Informed employees can act as a first line defence against cyber threats by identifying instances of social engineering like phishing and reporting it to management who can then inform other employees to be weary of additional phishing attempts. Moreover, informed employees can implement their own security measures to stay more secure against potential cyber threats and attacks. In addition, informed employees can also educate themselves on any recent cyber threats as well as suggest new procedures to protect the business from the new and advanced cyber threats.

Byte Solutions

Comprehensive Network Troubleshooting Guide

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Introduction

Ensuring a reliable wireless network for the business enables employees to perform their tasks with increased efficiency without any interruptions. Moreover, ensuring that the network is reliable in terms of security guarantees that the business complies with the relevant industry regulations and protect sensitive information from the business and any clients. The main aim of this troubleshooting guide is to provide information to the business's support team that can assist with resolving various network-related issues. This troubleshooting guide should offer solutions that can help identify issues that negatively impacts the overall performance of the network. In addition, this should also indicate methods that can solve any security concerns and vulnerabilities in the network.

Network overview

The components that make up the network infrastructure include access points, routers and switches. Access points refer to devices that enable devices such as laptops and IoT devices to wirelessly communicate with the network hardware. Routers are an important network device as it can be used to connect the business's network to other networks, like the internet, as well as manages the data traffic between networks. Switches are network devices that are used to manage traffic in the business network by utilising the MAC address of the destination of a data transfer, which improves the efficiency of data transfers.

For the Byte Solutions, access points and switches can be used to improve business performance by allowing employees to easily setup new devices that can communicate with the network as well as improve the data flow between wired devices. Routers can be used to enable devices that are connected through an access point or switch to easily communicate with each other if they are on different networks, which provides seamless operations between all smaller networks in the business.

Troubleshooting steps

Step 1: Initial assessment

Performing an initial assessment for the wireless network consists of analysing the wireless installation that looks at various configuration factors that can have an impact on the overall network performance and functionality (McNamee, 2019). A configuration factor that can impact the functionality of the network are the physical connections at the physical layer of the network. When issues occur at the physical layer it is best to ensure that all the cables are connected to the correct ports and don't show signs of any physical damage. The cable can be plugged into other ports or devices to ensure whether there is faulty network hardware or not. Moreover, swapping a cable that is suspected to be faulty with a good cable can be used to potentially resolve the issue. In addition, cable testers can be used to test the integrity of different cables to determine if it has been damaged.

Step 2: Connectivity issues

When experiencing issues related to the quality or lack of network connectivity, various issues can contribute to this such as attenuation, interference and authentication. Attenuation refers to the loss of signal strength due to cables exceeding their recommended distances, which is usually one hundred meters in length. Attenuation can be solved by incorporating either a repeater or a switch to boost signal strength to prevent it from getting lost or being weak. Interference is when transmission signals is disrupted due to external electrical sources like network equipment and fluorescent lights. Interference can be mitigated by utilising shielded twisted pair cables, which are designed to resist interference, or manage the cables in a manner that reduces the amount of interference that it experiences.

Authentication issues can arise due to various factors and can include incorrect user credentials and security protocol mismatches. If the employee is unable to remember their user credentials, then their credentials can be altered on the network that enables them to remember their credentials easier when needing to prove that they are authorised users. Additionally, if the security protocols of the user's device are not the same as the network, authentication issues will occur, and the user can solve this by changing their security settings to match with the network (portnox, 2024).

Step 3: Security issues

There are various methods that external threat actors can utilise to exploit vulnerabilities in the network, therefore it is important to identify these vulnerabilities and implement security features to mitigate them. Potential security related issues can be identified by performing vulnerability assessments and penetration tests to determine network vulnerabilities.

Unauthorised access refers to when attackers gain access to the network by using methods such as social engineering, password attacks and exploiting software vulnerabilities. Various methods to prevent unauthorised access includes implementing firewalls to filter both incoming and outgoing traffic to detect and block malicious traffic as well as ensuring that the software and firmware of the network is up to date with regular patches. Moreover, access control through methods like password policies and multi-factor authentication can be used to further mitigate the possibility for unauthorised users from gaining access to the network. In addition, encryption methods can be used to encrypt data when it is being transferred to prevent unauthorised users from intercepting the traffic and stealing sensitive information.

Step 4: Speed and performance

There are various methods that we can utilise to improve overall network speed and performance such as bandwidth testing and channel optimisation. Using bandwidth testing tools like iPerf and Speedtest.net enables us to monitor where instances of jittering and packet loss has occurred within the network. This should provide a general idea of what the performance issues in the network are and where it is happening, which allows us to easily develop a strategy to solve these issues. Moreover, channel optimisation refers to a set of tools or methods that can be used to improve the performance of the network. These methods that can be used include implementing a QoS(Quality of Service) that prioritises specified network traffic over other traffic, utilising load balancing across multiple networks to prevent devices from becoming a bottleneck and optimising how routing devices process TCP traffic (Petryschuk, 2024).

Conclusion

Any problems that are identified within the network such as security vulnerabilities, connectivity and performance issues should be addressed immediately to ensure that the network remains secure and efficient for the business operations. Furthermore, all factors that could be contributing to an issue should be inspected to ensure that the identified network problem is accurate and is solved effectively.

Ensuring that the network is maintained consistently can introduce various benefits to the business and the network. These benefits can include identifying opportunities for the business to reduce instances of downtime and increase the overall efficiency and performance of the network. Moreover, security vulnerabilities can also be identified and solved to improve the network's security and ensure that the business remains compliant with industry standards and regulations. In addition, constant network maintenance enables the business to perform less troubleshooting due to most issues being addressed through network maintenance.

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